

UNIVERSITY NAME

*probability and statistics — Final Exam**Year: 2021***Exercise 1:**

A parking lot has 10 spaces numbered from 1 to 10. The number of cars N arriving at the lot in one hour follows a Poisson distribution with parameter $\lambda > 0$. It is assumed that drivers choose a space at random and independently of one another.

Let G be the event “the driver chooses space number 1” and X be the random variable “number of drivers who choose space number 1”.

Calculate:

1. $E(N)$
2. $P(G)$
3. $P(X = k \mid N = m)$ (distinguish the cases $k > m$ and $k \leq m$)
4. $P(X = k)$
5. $E(X)$ and $\text{Var}(X)$
6. The probability generating function of N and that of X .

Answer Area

1. $E(N)$

$N \sim \mathcal{P}(\lambda)$, so:

$$E(N) = \lambda.$$

2. $P(G)$

Each driver chooses a space uniformly among 10 spaces:

$$P(G) = \frac{1}{10}.$$

3. $P(X = k \mid N = m)$

Given $N = m$, drivers choose independently. $X \mid N = m \sim \text{Binomial}(m, \frac{1}{10})$.
Thus:

$$P(X = k \mid N = m) = \begin{cases} \binom{m}{k} \left(\frac{1}{10}\right)^k \left(\frac{9}{10}\right)^{m-k} & \text{if } 0 \leq k \leq m, \\ 0 & \text{if } k > m. \end{cases}$$

4. $P(X = k)$

By total probability:

$$\begin{aligned} P(X = k) &= \sum_{m=k}^{\infty} P(X = k \mid N = m)P(N = m) \\ &= \sum_{m=k}^{\infty} \binom{m}{k} \left(\frac{1}{10}\right)^k \left(\frac{9}{10}\right)^{m-k} \cdot \frac{e^{-\lambda} \lambda^m}{m!}. \end{aligned}$$

Let $j = m - k$, then $m = k + j$:

$$P(X = k) = e^{-\lambda} \left(\frac{1}{10}\right)^k \sum_{j=0}^{\infty} \binom{k+j}{k} \left(\frac{9}{10}\right)^j \frac{\lambda^{k+j}}{(k+j)!}.$$

Note: $\binom{k+j}{k} \frac{1}{(k+j)!} = \frac{1}{k!j!}$. So:

$$P(X = k) = e^{-\lambda} \frac{(\lambda/10)^k}{k!} \sum_{j=0}^{\infty} \frac{(9\lambda/10)^j}{j!} = e^{-\lambda} \frac{(\lambda/10)^k}{k!} e^{9\lambda/10}.$$

Thus:

$$P(X = k) = \frac{e^{-\lambda/10} (\lambda/10)^k}{k!}.$$

5. $E(X)$ and $\text{Var}(X)$

Since $X \sim \mathcal{P}(\lambda/10)$:

$$E(X) = \frac{\lambda}{10}, \quad \text{Var}(X) = \frac{\lambda}{10}.$$

6. Probability generating functions

For N : $G_N(s) = E(s^N) = e^{\lambda(s-1)}$.

For X : Since $X \sim \mathcal{P}(\lambda/10)$, $G_X(s) = e^{(\lambda/10)(s-1)}$.

Exercise 2:

Let X be a uniformly distributed random variable on the interval $[-1, 1]$.

1. Find the distribution function of the random variable $Y = 4 - X^2$.
2. Compute $P(2 \leq Y \leq 3.5)$.

Answer Area

$X \sim \mathcal{U}([-1, 1])$, so $f_X(x) = \frac{1}{2} \mathbf{1}_{[-1, 1]}(x)$.

1. Distribution function of $Y = 4 - X^2$

Note: Y takes values in $[3, 4]$, since $X^2 \in [0, 1]$. For $y \in [3, 4]$:

$$F_Y(y) = P(Y \leq y) = P(4 - X^2 \leq y) = P(X^2 \geq 4 - y).$$

Let $t = 4 - y \in [0, 1]$. Then:

$$F_Y(y) = P(X^2 \geq t) = P(|X| \geq \sqrt{t}) = 1 - P(|X| < \sqrt{t}).$$

Since f_X is symmetric:

$$P(|X| < \sqrt{t}) = \int_{-\sqrt{t}}^{\sqrt{t}} \frac{1}{2} dx = \frac{1}{2} \cdot 2\sqrt{t} = \sqrt{t}.$$

Thus:

$$F_Y(y) = 1 - \sqrt{4 - y}, \quad y \in [3, 4].$$

Outside: $F_Y(y) = 0$ for $y < 3$, and $F_Y(y) = 1$ for $y > 4$.

The density: for $y \in (3, 4)$:

$$f_Y(y) = \frac{d}{dy} F_Y(y) = \frac{d}{dy} (1 - (4 - y)^{1/2}) = \frac{1}{2} (4 - y)^{-1/2}.$$

Thus:

$$F_Y(y) = \begin{cases} 0 & y < 3, \\ 1 - \sqrt{4 - y} & 3 \leq y \leq 4, \\ 1 & y > 4, \end{cases} \quad f_Y(y) = \begin{cases} \frac{1}{2\sqrt{4-y}} & 3 < y < 4, \\ 0 & \text{otherwise.} \end{cases}$$

2. Compute $P(2 \leq Y \leq 3.5)$

Since $Y \geq 3$, this is $P(3 \leq Y \leq 3.5)$. Using the CDF:

$$P(3 \leq Y \leq 3.5) = F_Y(3.5) - F_Y(3) = (1 - \sqrt{4 - 3.5}) - (1 - \sqrt{4 - 3}).$$

Compute: $4 - 3.5 = 0.5$, $\sqrt{0.5} = \frac{\sqrt{2}}{2}$. $4 - 3 = 1$, $\sqrt{1} = 1$. So:

$$P = \left(1 - \frac{\sqrt{2}}{2}\right) - (1 - 1) = 1 - \frac{\sqrt{2}}{2} - 0 = 1 - \frac{\sqrt{2}}{2}.$$